How Sleep Duration Affects Stress Levels in College Students

Husna Manalai
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Background:

Sleep is an important part of student health because it affects mood, concentration, and overall well-being. College students are a great group to study because many students have irregular sleep schedules due to academic work, social life, jobs, and stress. Previous research has found that sleep problems are related to mental health problems in college students. For example, Milojevich and Lukowski found that lack of sleep was associated with mental health in undergraduate students, even among students who generally had healthy sleep habits. [1] Other studies have also found that poor sleep or insomnia is associated with stress, anxiety, depression, or other mental health concerns among college students.[2]

The purpose of this project is to examine whether sleep duration is related to stress levels in college students. My original hypothesis was that students who sleep less would report higher stress scores than students who sleep 7 or more hours. I also thoughts that there would be a negative correlation between sleep hours and stress meaning that as sleep hours increased stress scores would decrease.

Methods:

For this project I used two datasets provided for the final project. The first dataset included 20 observations with two variables: sleep hours and stress score. The second dataset included stress scores for three different sleep duration groups: students who slept less than 7 hours, students who slept 7-8 hours, and students who slept more than 8 hours.

For the first dataset, I used an independent samples t-test to compare the mean stress score of students who slept less than 7 hours with the mean stress score of students who slept 7 or more hours. I used this test because I was comparing the means of two independent groups. The null hypothesis was that there would be no difference in mean stress score between the two groups. The alternative hypothesis was that the two groups would have different mean stress scores.

I also used a Pearson correlation test to examine the relationship between sleep hours and stress score as continuous variables. I used this test because I wanted to see whether there was a linear relationship between the number of hours slept and stress score. The null hypothesis was that there would be no relationship between sleep hours and stress score. The alternative hypothesis was that sleep hours and stress score would be related.

For the second dataset, I used a one-way ANOVA to compare stress scores across three sleep duration groups. I used this test because there were three group means being compared. The null hypothesis was that all three sleep groups would have the same mean stress score. The alternative hypothesis was that at least one sleep group would have a different mean stress score.

For the t-test, I calculated the means, sample standard deviations, variance, estimated standard error, and t statistic. For the correlation, I calculated SSx, SSy, SSxy, and Pearson's r. For the ANOVA, I calculated the between-groups sum of squares, within-groups sum of squares, degrees of freedom, mean squares, and F ratio.

Results:

For the first dataset, I divided students into two groups: students who slept less than 7 hours and students who slept 7 or more hours. The less than 7 hours group had 9 students, with a mean stress score of 5.60 and a standard deviation of 1.64. The 7 or more hours group had 11 students, with a mean stress score of 7.70 and a standard deviation of 1.53.

An independent samples t-test showed that this difference was statistically significant, $t(18) = -2.96$, $p = 0.0084$. However, the direction of the result was opposite of my original hypothesis. Students who slept 7 or more hours had higher stress scores than students who slept less than 7 hours.

The t-test summary table is shown below:

```
INDEPENDENT SAMPLES T-TEST
--------------------------
Less than 7 hours: n = 9 M = 5.603 SD = 1.639
7 or more hours:  n = 11 M = 7.704 SD = 1.529
df = 18
pooled variance = 2.492
SEM = 0.71
t = -2.96
Cohen's d = -1.331
```

I also used a Pearson correlation test to examine the relationship between sleep hours and stress score. The correlation was $r = 0.618$, with $r^2 = 0.381$. This means that about 38.1% of the variation in stress scores was associated with sleep hours in this dataset. The correlation was statistically significant, $p = 0.0037$. This was a positive correlation, meaning that higher sleep duration was associated with higher stress scores in this dataset.

Figure 1 shows the relationship between sleep hours and stress score. Each point represents one student. The scatterplot shows a positive trend, which matches the positive correlation.

The summary table is shown below:

```
PEARSON CORRELATION
-------------------
SSx = 54.799
SSy = 66.705
SSxy = 37.343
r = 0.618
r squared = 0.381
df = 18
```

For the second dataset, I compared stress scores across three sleep duration groups. Students who slept less than 7 hours had the highest mean stress score, with a mean of 6.76 and a standard deviation of 1.07. Students who slept 7–8 hours had a mean stress score of 5.33 and a standard deviation of 1.77. Students who slept more than 8 hours had a mean stress score of 5.54 and a standard deviation of 1.29.

Figure 2 shows the mean stress score for each sleep duration group. Students who slept less than 7 hours had the highest average stress score.

A one-way ANOVA showed that there was a statistically significant difference among the three sleep groups, $F(2,72) = 7.58$, $p = 0.0010$. The effect size was $\eta^2 = 0.174$, which is considered a large effect using the class guideline that .14 is large.

The ANOVA summary table is shown below:

```
ONE-WAY ANOVA
-------------
< 7 hours : n = 25 M = 6.764 SD = 1.069
7-8 hours : n = 25 M = 5.332 SD = 1.767
> 8 hours : n = 25 M = 5.536 SD = 1.294

Source       SS       df  MS       F
Between       30.002   2   15.001   7.579
Within        142.51   72  1.979
Total         172.511      74
eta squared = 0.174
```

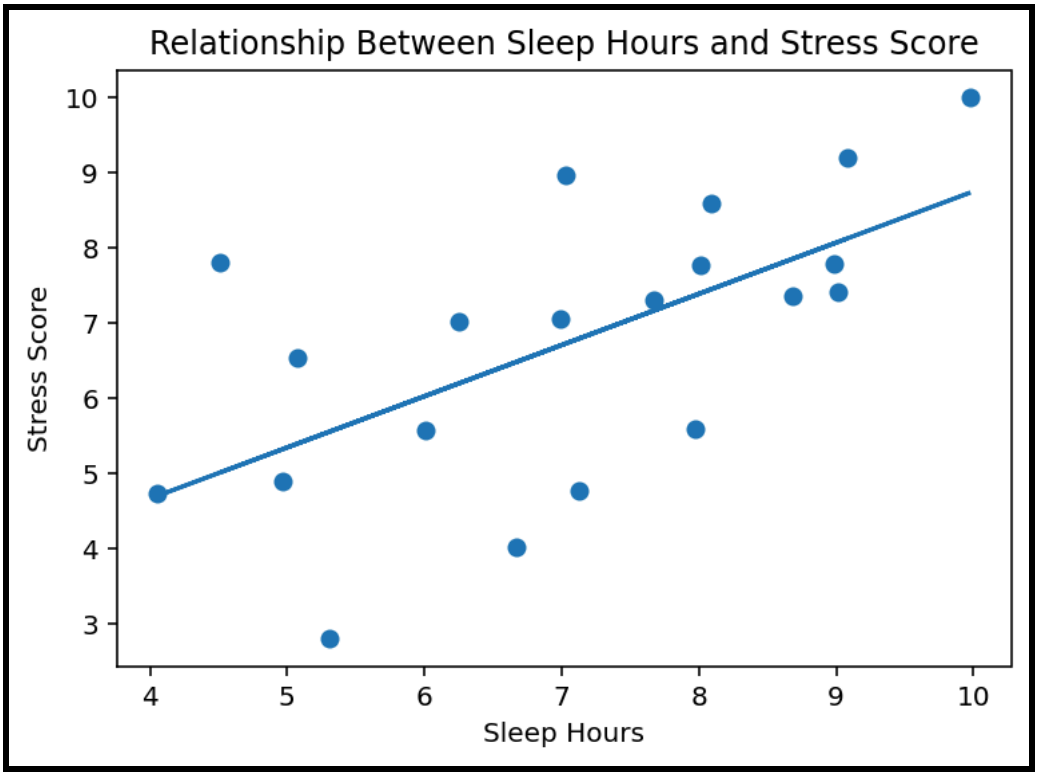


Figure 1: Scatterplot showing the relationship between sleep hours and stress score. Each point represents one student.

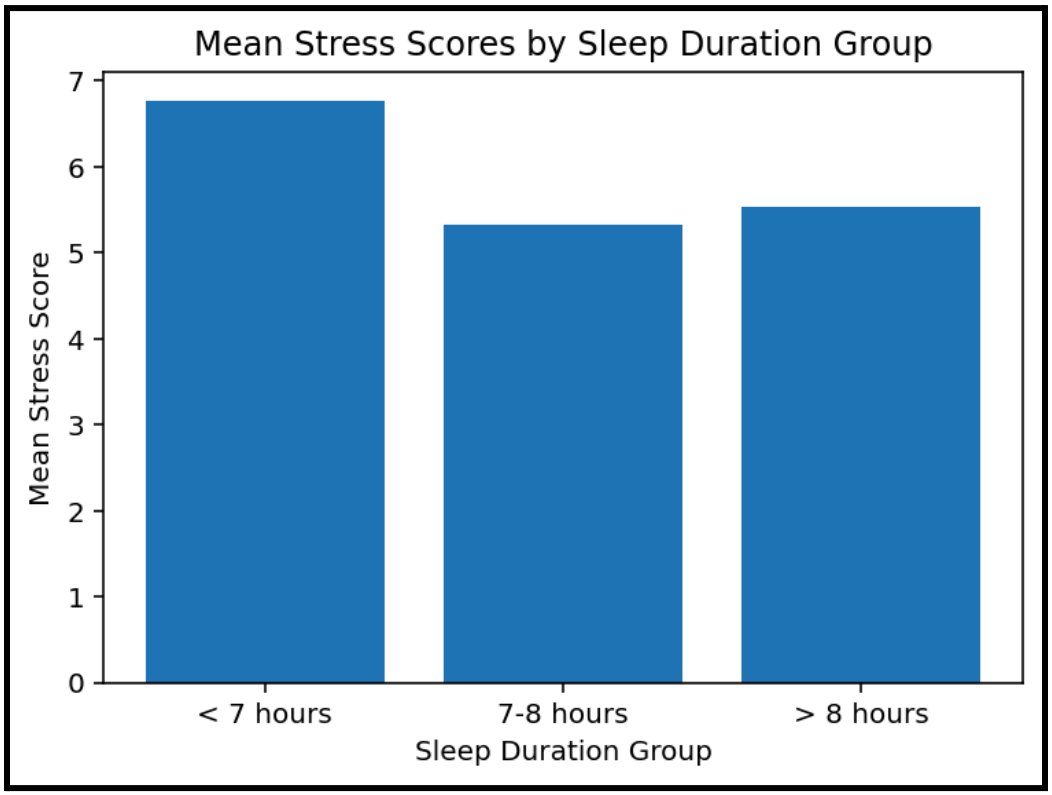


Figure 2: Mean stress scores by sleep duration group.

Discussion:

The results were mixed. In the first dataset, students who slept more had higher stress scores, which was the opposite of my original hypothesis. The Pearson correlation was also positive, meaning that more sleep was associated with higher stress in this dataset. This was surprising because I expected students with less sleep to have higher stress.

One possible explanation is that the first dataset had a small sample size of only 20 students. With a small sample, a few unusual values can have a strong effect on the results. Another possible explanation is that sleep duration does not tell us the full story about sleep. A student may sleep for many hours but still have poor sleep quality, wake up often, or feel tired during the day. In that case, longer sleep duration would not necessarily mean better rest.

It is also possible that the direction of the relationship is more complicated. Instead of low sleep causing stress, students who are already highly stressed may sleep more because they feel exhausted or burned out. Stress could also affect sleep patterns differently for different students. Some students may sleep less when stressed, while others may sleep more.

The second dataset supported my original hypothesis more clearly. Students who slept less than 7 hours had the highest average stress score, and the ANOVA showed that there was a statistically significant difference among the three groups. This result fits better with previous research showing that poor sleep and sleep problems are related to worse mental health outcomes in college students [1, 2].

Overall, these results suggest that sleep and stress are related, but the relationship is not simple. Sleep duration alone may not be enough to explain stress levels. In future I will also include sleep quality, anxiety, depression, academic workload, caffeine use, and time of semester. Therefore, while the results do not all support my original hypothesis, they show that sleep duration is related to stress and that this relationship may depend on other factors besides the number of hours slept.

Acknowledgements:

I used the datasets provided by my professor for this final project. I also used formulas and examples from Brightspace and used the codes from the labs we've done in class to calculate the t-test, Pearson correlation, and one-way ANOVA. I used Python to organize the data, calculate statistics, and create the figures.

References:

[1] Milojevich, H. M., & Lukowski, A. F. (2016). Sleep and mental health in undergraduate students with generally healthy sleep habits. PLOS ONE, 11(6), e0156372.

[2] Mbous, Y. P. V., et al. (2022). Psychosocial correlates of insomnia among college students. Preventing Chronic Disease, 19.